IE 7275 Data Mining in Engineering
Homework 3
Deadline: 11/05/2022

Note:

- All the files need to be submitted along with your solution sheets on Canvas.
- Use Python or equivalent programing languages ONLY.
- Specify your implementation steps clearly.

Problem 1 (Logistic Regression) [15 points]

The file Banks.csv includes data on a sample of 20 banks. The "Financial Condition" column records the judgment of an expert on the financial condition of each bank. This outcome variable takes one of two possible values—weak or strong—according to the financial condition of the bank. The predictors are two ratios used in the financial analysis of banks: TotLns&Lses/Assets is the ratio of total loans and leases to total assets and TotExp/Assets is the ratio of total expenses to total assets. The target is to use the two ratios for classifying the financial condition of a new bank.

(a) Using Python, develop a logistic regression model (on the entire dataset) that models the status of a bank as a function of the two financial measures provided. Specify the success class as weak (this is similar to creating a dummy that is 1 for financially weak banks and 0 otherwise).

(b) Write the estimated equation that associates the financial condition of a bank with its two predictors in three formats:

1. The logit as a function of the predictors.
2. The odds as a function of the predictors
3. The probability as a function of the predictors

(c) Consider a new bank whose total loans and leases/assets ratio = 0.6 and total expenses/assets ratio = 0.11. From your logistic regression model, estimate the following four quantities for this bank (use Python to do all the intermediate calculations; show your final answers to four decimal places): the logit, the odds, the probability of being financially weak, and the classification of the bank (use cutoff = 0.5).

(d) The cutoff value of 0.5 is used in conjunction with the probability of being financially weak. Compute the threshold that should be used if we want to make a classification based on the odds of being financially weak, and the threshold for the corresponding logit.

(e) Interpret the estimated coefficient for the total loans & leases to total assets ratio (TotLns&Lses/Assets) in terms of the odds of being financially weak.

(f) When a bank that is in poor financial condition is misclassified as financially strong, the misclassification cost is much higher than when a financially strong bank is misclassified

as weak. To minimize the expected cost of misclassification, should the cutoff value for classification (which is currently at 0.5) be increased or decreased?

Problem 2 (Logistic Regression) [10 points]

Table 3 shows the information of logistic regression trained with data of two variables: experience and training, to predict if staff can complete tasks on time.

(a) Write down the logit function of variables. Construct a probability function of of staff completing tasks on time (Y=1).

(b) Interpret the impact of the two variables on prediction outcomes. Hint: odd ratios.

(c) Predict three new staff members of records [9, 8], [9, 7], and [8, 7]. Show the probability of staff completing tasks on time (Y=1).

Table 1. for Problem 2.

Input Variables	Coefficient	Std. Error	Chi2-Statistic	P-Value
Intercept	-10.98	2.89	14.42	0.00
Experience	1.13	0.29	15.01	0.00
Training	0.18	0.34	0.28	0.59

Problem 3 (Linear Discriminant Analysis) [20 points]

Universal Bank is a relatively young bank growing rapidly in terms of overall customer acquisition. The majority of these customers are liability customers with varying sizes of relationship with the bank. The customer base of asset customers is quite small, and the bank is interested in expanding this base rapidly to bring in more loan business. In particular, it wants to explore ways of converting its liability customers to personal loan customers.

A campaign the bank ran for liability customers last year showed a healthy conversion rate of over 9% successes. This has encouraged the retail marketing department to devise smarter campaigns with better target marketing. The goal of our analysis is to model the previous campaign's customer behavior to analyze what combination of factors make a customer more likely to accept a personal loan. This will serve as the basis for the design of a new campaign.

The file UniversalBank.csv contains data on 5000 customers. The data include customer demographic information (e.g., age, income), the customer's relationship with the bank (e.g., mortgage, securities account), and the customer response to the last personal loan campaign (Personal Loan). Among these 5000 customers, only 480 (= 9.6%) accepted the personal loan that was offered to them in the previous campaign.

Using Python, partition the data (60% training and 40% validation) and then perform a discriminant analysis that models Personal Loan as a function of the remaining predictors (excluding zip code). Remember to turn categorical predictors with more than two categories into dummy variables first. Specify the success class as 1 (personal loan acceptance), and use the default cutoff value of 0.5.

(a) Compute summary statistics for the predictors separately for loan acceptors and nonacceptors. For continuous predictors, compute the mean and standard deviation. For

categorical predictors, compute the percentages. Are there predictors where the two classes differ substantially?

(b) Examine the model performance on the validation set.
   1. What is the accuracy rate?
   2. Is one type of misclassification more likely than the other?
   3. Select three customers who were misclassified as acceptors and three who were misclassified as nonacceptors. The goal is to determine why they are misclassified. First, examine their probability of being classified as acceptors: is it close to the threshold of 0.5? If not, compare their predictor values to the summary statistics of the two classes to determine why they were misclassified.

(c) As in many marketing campaigns, it is more important to identify customers who will accept the offer rather than customers who will not accept it. Therefore, a good model should be especially accurate at detecting acceptors. Examine the lift chart and decile-wise lift chart for the validation set and interpret them in light of this ranking goal.

(d) Compare the results from the discriminant analysis with those from a logistic regression (both with cutoff 0.5 and the same predictors). Examine the confusion matrices, the lift charts, and the decile charts. Which method performs better on your validation set in detecting the acceptors?

(e) The bank is planning to continue its campaign by sending its offer to 1000 additional customers. Suppose that the cost of sending the offer is $1 and the profit from an accepted offer is $50. What is the expected profitability of this campaign?

(f) The cost of misclassifying a loan acceptor customer as a nonacceptor is much higher than the opposite misclassification cost. To minimize the expected cost of misclassification, should the cutoff value for classification (which is currently at 0.5) be increased or decreased?

Problem 4 (Linear Discriminant Analysis) [20 points]

Consider the data in Table 2. Answer the following questions:

(a) Compute $\boldsymbol{\mu}_{+1}$ and $\boldsymbol{\mu}_{-1}$, and $\mathbf{S}_B$, the between-class scatter matrix.

(b) Compute $\mathbf{S}_{+1}$ and $\mathbf{S}_{-1}$, and $\mathbf{S}_W$, the within-class scatter matrix.

(c) Find the best direction $\mathbf{w}$ that discriminates between the classes. Use the fact that the inverse of the matrix $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is given as $\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

(d) Having found the direction $\mathbf{w}$, find the point on $\mathbf{w}$ that best separates the two classes.

Note: you need to elaborate all details.

Problem 5 (SVM) [15 points]

Consider the training dataset in Table 3.

(a) Solve the problem using Python package or equivalent programming language and provide the optimal hyperplane and support vectors in a SVM classifier.

(b) Write down the soft-margin SVM model.
(c) Check the optimality using KKT conditions.
(d) Draw all data points and the hyperplane on a 2-dimensional space and highlight the support vectors.

Problem 6 (SVM) [20 points]

Consider a dataset of Personal Loan Acceptance (1500 records)

(a) Using Python, show the SVM classification function from 1000 training records. Specify your setting in building a SVM model.
(b) What are the support vectors?
(c) In Python, apply a 5-fold cross validation and report the average training and testing accuracy, respectively.

Table 2. for Question 3.

	x1	x2	Class
1	0.54	0.78	1
2	0.78	0.71	1
3	0.59	0.92	1
4	0.56	0.88	1
5	0.79	1.00	1
6	1.00	0.81	1
7	0.98	0.75	1
8	0.75	0.95	1
9	0.63	0.85	1
10	0.84	0.88	1
11	0.46	0.93	1
12	0.74	0.85	1
13	0.68	0.83	0
14	0.48	0.88	0
15	0.59	0.73	0
16	0.39	0.86	0
17	0.76	0.75	0
18	0.45	0.75	0
19	0.54	0.68	0
20	0.60	0.78	0
21	0.43	0.69	0
22	0.30	0.80	0
23	0.46	0.59	0
24	0.57	0.63	0

Table 3. for Problem 5.

A	B	Class
3.5	4.0	1
2.0	4.0	1
2.0	6.0	1
1.5	7.0	1
7.0	6.5	1
2.1	2.5	0
8.0	4.0	0
9.1	4.5	0